Hi, I'm Xin Jie!



> Hi! I'm Fung Xin Jie, currently pursuing an *Engineering Systems and Design Degree* at the Singapore University of Technology and Design (SUTD) with a *minor in Design, Technology and Society.* Prior to this, I earned a *Diploma in Applied Chemistry* from Singapore Polytechnic.
>
> My interdisciplinary education allows me to approach problems from both technical and human-centered perspectives, combining systems thinking, analytical reasoning, and design awareness to develop practical and meaningful solutions. Having grown up overseas, I also learned to adapt quickly to unfamiliar environments, think on my feet, and collaborate with people from diverse backgrounds.
>
> I'm particularly interested in **data analytics**, **systems optimisation**, and **leveraging technology** to improve decision-making, operations, and user experiences.

About Me

Growing up overseas (Kazakhstan & US) exposed me to different cultures, environments, and ways of thinking from a young age. Constantly adapting to new situations taught me how to think quickly on my feet, stay open-minded, and approach challenges with curiosity rather than hesitation. These experiences shaped the way I work today: adaptable, observant, and comfortable navigating uncertainty.

My academic journey began in *Applied Chemistry at Singapore Polytechnic (SP)*, where I developed a strong foundation in analytical thinking, problem-solving, and experimentation. In SP, I took an introductory programming elective in Python, which sparked my interest in technology and data-driven problem solving. What started as curiosity quickly grew into a passion for understanding how technology, systems, and data can be used to solve real-world challenges. This experience ultimately inspired me to pivot into the tech field and pursue *Engineering Systems and Design at SUTD.*

This interest deepened further during my internship at *Procter & Gamble as an Associate Researcher* under the *Product Research* function. The role pushed me beyond my traditional chemistry background and challenged me to adapt quickly in a fast-paced, interdisciplinary environment. Learning on the job taught me how science, technology, consumer insights, and business strategy intersect to drive innovation and decision-making.

What excites me most is solving complex problems that sit at the intersection of data, systems, and human behaviour. I enjoy breaking down messy challenges, identifying patterns, and finding ways to improve processes and decision-making. Whether through analytics, systems thinking, or design, I'm motivated by work that creates meaningful and practical impact.

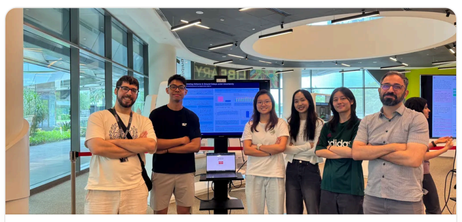
Outside of work, I'm drawn to creative pursuits: writing, reading widely across genres, and watching films. I find that engaging with stories and different perspectives keeps me curious and often informs how I approach problems in unexpected ways.

Selected Projects



Synthetic Data Generator for Credit Bureau Singapore

Business Analytics · Python · SQL · AI



Optimising Airborne and Ground Delays under Uncertainty

Optimisation · Python · SQL · AI



Design Thinking and Innovation: Plantbot

Hardware · UX/Design · Python



Recycling Collection Routing Problem: Multi-Trip Vehicle Routing with Depot-Based Unloading

Optimisation · Python · Metaheuristic Algorithms · AI

Associate Researcher Intern with Procter & Gamble

User Testing · R&D · Analytical Chemsitry

Experience

Data Analyst Intern - Sustainability @ *The Ascott Limited*

Sept 2026 - Present

Working with the Sustainability department on environmental and operational data monitored across Ascott's global portfolio of serviced residences. My focus is on data consolidation, quality checks, and reporting to support ESG tracking and disclosure across markets, with an emphasis on streamlining and optimising how sustainability data is processed and visualised across properties.

Private Tuition Teacher - Primary School English & Secondary School Science

May 2024 - June 2026

I provided one-on-one tuition across Primary English, Secondary 1 Science, and Secondary 4 Science. Beyond subject knowledge, I focus on building each student's confidence and genuine understanding of the material. One of my students achieved an A2 in their preliminary exams within less than two months of working together.

Associate Researcher Intern - Air Care, Research and Development @ *Procter & Gamble*

March 2023 - August 2023

As part of the R&D team working on Febreze and Ambi Pur, I created product demonstrations, led in-house usability testing, and independently planned and executed research projects across different brands. The role required strong cross-functional collaboration, working with different departments (Analytical, Microbial, Marketing) to meet project goals, and the ability to manage diverse creative needs in a fast-paced environment.

✨ Relevant Courses

Data and Business Analytics: A client-facing course where my team worked with Credit Bureau Singapore to identify system improvements using real data, helping them optimise their data generation and anonymisation system. Covered the full analytics pipeline including data manipulation, visualisation, analysis, and time series forecasting, with findings delivered as oral, written, and poster presentations.

Optimisation: An introduction to the mathematical foundations of decision-making, covering linear programming, network optimisation, integer programming, and dynamic programming applied to real-world problems in resource allocation, logistics, and system design.

Heuristics and Systems Theory: A follow-on to Optimisation covering metaheuristic algorithms (simulated annealing, evolutionary algorithms, swarm optimisation) and dynamical systems theory for modelling and control. Implemented algorithms hands-on in Python.

The Analytics Edge: A practical course applying regression, classification trees, clustering, and text analytics to real-world cases in business, healthcare, and sport. Used R to build and evaluate models with a focus on turning data into actionable decisions.

⭐ Skills

Technical
Python, SQL, Julia, R, Gurobi, Microsoft Excel

Data & Analytics
Data manipulation, data visualisation, regression analysis, time series forecasting, machine learning

Design & Creative
Adobe Lightroom, Adobe Photoshop, Canva

Research & Lab
Analytical chemistry, FTIR spectroscopy, gas chromatography (GC), UV-Vis spectroscopy, experimental design, data collection and analysis

Soft Skills
Collaboration across cross-functional and interdisciplinary teams, project management, client-facing communication, oral and written presentation, human-centred design thinking, teaching and mentoring

📬 Contact

Reach me at *fung.xj@gmail.com* or on LinkedIn at *www.linkedin.com/in/xinjiefung* !

My creative writing blog:

- Station XJ